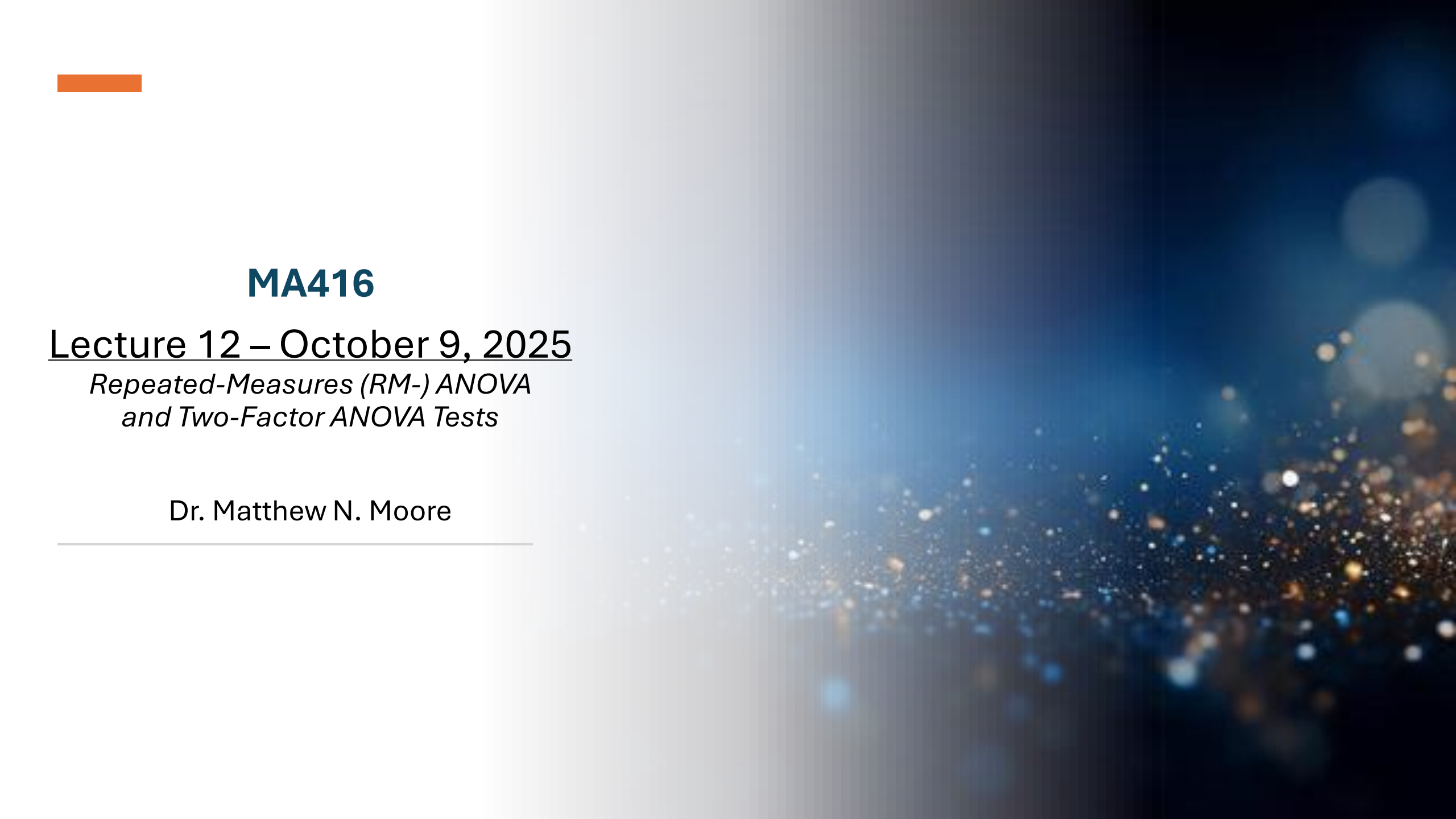
MA416

Lecture 12 – October 9, 2025

Repeated-Measures (RM-) ANOVA and Two-Factor ANOVA Tests

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Review on RM-ANOVA Assumptions

Example: Do we have evidence to believe that there is a significant difference in the average heights over three different periods of time?

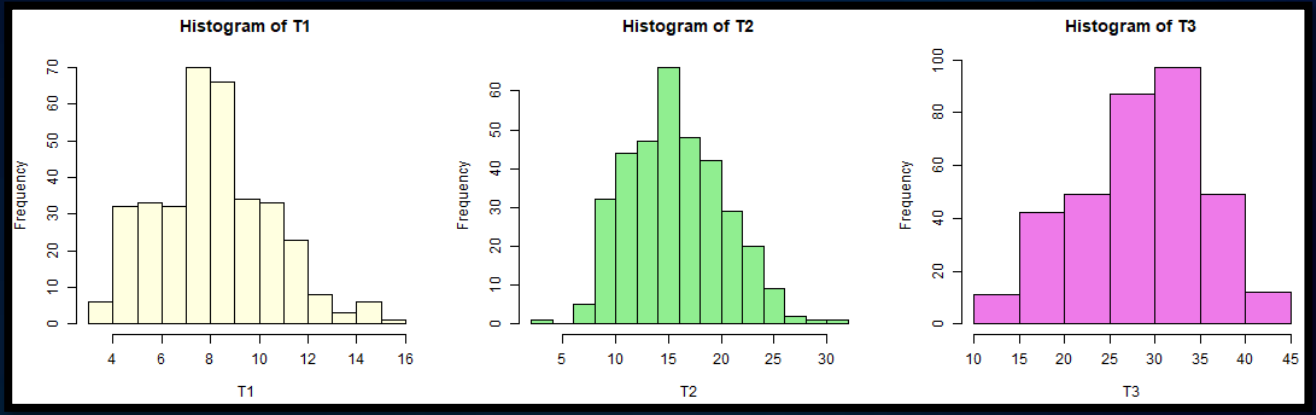
RM-ANOVA Assumptions

1. Normality of Groups
2. Compound Symmetry Between Groups
3. Independence Within Groups

```
mydata = read.csv('FlowerGrowth_9.25.csv')

T1 = mydata$Height_2Weeks_cm
T2 = mydata$Height_1Month_cm
T3 = mydata$Height_2Months_cm

par(mfrow=c(1,3))
hist(T1, col='lightyellow')
hist(T2, col='lightgreen')
hist(T3, col='orchid2')
```




```
options(digits = 5)
source('mymauchly.R') # Try to build your own w/o AI

MM = mymauchly(cbind(T1, T2, T3), alpha=0.05)

MM$W
c(MM$mpval_o, MM$mpval_m)
```

```
> MM$W
[1] 0.10158
> c(MM$mpval_o, MM$mpval_m)
[1] 3.4100e-171 5.7605e-172
```

RM-ANOVA Test

We first consider the case where all assumptions are satisfied. Suppose that x_{ij} is the ith person in the jth group, with $i = 1,2, \dots, n_j$ and $j = 1,2, \dots, j$, with (ideally) $n_1 = n_2 = \cdots = n_g \equiv n$. Then:

$$SST = \sum_{i=1}^{n}\sum_{j=1}^{g}\left(x_{ij} - \bar{\bar{x}}\right)^2 = \cdots = s_{all}^2 \cdot (ng - 1)$$

Also, like ANOVA, we have:

$$SSM = \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{\bar{x}}\right)^2 = n \cdot \sum_{j=1}^{g}\left(\bar{x}_j - \bar{\bar{x}}\right)^2$$

Unlike the ANOVA, we have the variance-within-subjects term:

$$SSS = \sum_{i=1}^{n} g \cdot (\bar{x}_i - \bar{\bar{x}})^2 = g \cdot \sum_{i=1}^{n}(\bar{x}_i - \bar{\bar{x}})^2$$

RM-ANOVA Test

$$SST = \sum_{i=1}^{n}\sum_{j=1}^{g}\left(x_{ij} - \bar{\bar{x}}\right)^2 = \cdots = s_{all}^2 \cdot (ng-1)$$

$$SSM = \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{\bar{x}}\right)^2 = n \cdot \sum_{j=1}^{g}\left(\bar{x}_j - \bar{\bar{x}}\right)^2$$

$$SSS = \sum_{i=1}^{n} g \cdot (\bar{x}_i - \bar{\bar{x}})^2 = g \cdot \sum_{i=1}^{n}(\bar{x}_i - \bar{\bar{x}})^2$$

We also have the usual (but differently defined from ANOVA) as:

$$SSE = \sum_{i=1}^{n}\sum_{j=1}^{g}\left(x_{ij} - \bar{x}_i - \bar{x}_j + \bar{\bar{x}}\right)^2$$

```
X = cbind(T1, T2, T3)

xbarj = colMeans(X)
xbari = rowMeans(X)
xbarbar = mean(X)
s2all = var(c(T1, T2, T3))

n = length(T1)
g = 3

SST = s2all*(n*g-1)
SSM = n*sum((xbarj - xbarbar)^2)
SSS = g*sum((xbari - xbarbar)^2)

E = X - xbari - matrix(xbarj, n, g, byrow=T) + xbarbar
SSE = sum(E^2)
```

Question: Does $SST = SSE + SSS + SSM$?

```
SST
SSM+SSS+SSE
```



```
> SST
[1] 98709
> SSM+SSS+SSE
[1] 98709
```

RM-ANOVA Test

$$SST = \sum_{i=1}^{n}\sum_{j=1}^{g}\left(x_{ij} - \bar{\bar{x}}\right)^2 = \cdots = s_{all}^2 \cdot (ng-1)$$

$$SSM = \sum_{j=1}^{g} n_j \left(\bar{x}_j - \bar{\bar{x}}\right)^2 = n \cdot \sum_{j=1}^{g}\left(\bar{x}_j - \bar{\bar{x}}\right)^2$$

$$SSS = \sum_{i=1}^{n} g \cdot (\bar{x}_i - \bar{\bar{x}})^2 = g \cdot \sum_{i=1}^{n} (\bar{x}_i - \bar{\bar{x}})^2$$

$$SSE = \sum_{i=1}^{n}\sum_{j=1}^{g}\left(x_{ij} - \bar{x}_i - \bar{x}_j + \bar{\bar{x}}\right)^2$$

Metric	Degrees of Freedom	Mean-Square Metrics
SST	$ng-1$	$\frac{SST}{ng-1}$
SSS	$n-1$	$\frac{SSS}{n-1}$
SSM	$g-1$	$\frac{SSM}{g-1}$
SSE	$ng-n-g+1$	$\frac{SSE}{(n-1)(g-1)}$

Null Hypothesis:

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_g \quad \rightarrow \quad F_{stat} := \frac{MSM}{MSE} \sim F_{[g-1,(n-1)(g-1)]}$$

```
> Fstat = MSM/MSE
> Fstat
[1] 2989.3
```

Non-Spherical RM-ANOVA Test

Question: What if sphericity fails? Can we "fix" the RM-ANOVA like we "fixed" ANOVA when homoscedasticity wasn't met?

Null Hypothesis:

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_g \qquad \rightarrow \qquad F_{stat} \coloneqq \frac{MSM}{MSE} \sim F_{[\nu_m, \nu_e]}$$

where:

$$\nu_m = \varepsilon \cdot (g-1) \qquad \text{and} \qquad \nu_e = \varepsilon \cdot (n-1)(g-1)$$

Ideally, if $W \coloneqq \frac{\det(\hat{\Sigma}_x)}{\left[\frac{\mathrm{tr}(\hat{\Sigma}_x)}{g}\right]^g} \approx 1$, then we want $\varepsilon \approx 1$, and if $W \ll 1$, then we want $0 < \varepsilon \ll 1$.

Two common approaches:

1. If sphericity violation is mild/moderate, use ε_{HF} (Huynh-Feldt)
2. If sphericity violation is severe, use ε_{GG} (Greenhouse-Geisser)

Non-Spherical RM-ANOVA Test

Two common approaches:

1. If sphericity violation is mild/moderate, use ε_{HF} (Huynh-Feldt)
2. If sphericity violation is severe, use ε_{GG} (Greenhouse-Geisser)

Note: Sphericity (tied to variance) is the issue; regardless of group/individual/grand mean values.

Define:

$$C = \left(I_{g\times g}\right) - \frac{1}{g}\cdot\left(1_{g\times g}\right) \qquad \rightarrow \qquad \hat{\Sigma}_c = C\cdot\hat{\Sigma}_x\cdot C$$

```
sighatx = var(X)
ones = matrix(1, nrow=g, ncol=g)
C = diag(g) - (1/g)*ones
sighatc = C%*%sighatx%*%C
```

Our correction factors are then given by:

$$\varepsilon_{GG} = \frac{\left[\mathrm{tr}(\hat{\Sigma}_c)\right]^2}{(g-1)\cdot\mathrm{tr}\left\{\left(\hat{\Sigma}_c\right)^2\right\}} \qquad \text{and} \qquad \varepsilon_{HF} = \frac{n(g-1)\varepsilon_{GG} - 2}{(g-1)(n-1) - (g-1)\varepsilon_{GG}}$$

```
> eps_gg
[1] 0.78406
> eps_hf
[1] 0.78521
```

```
> v_msm = (g-1)*c(eps_hf, eps_gg)
> v_mse = (n-1)*(g-1)*c(eps_hf, eps_gg)
> v_msm
[1] 1.5704 1.5681
> v_mse
[1] 543.37 542.57
```

Common Practice:

$$0 < \frac{1}{g-1} \leq^* \varepsilon_* \leq^* 1 < \infty$$

Two-Factor ANOVA

Goal: To assess whether there is a significant difference in the average values between multiple groups, where those groups are generated from one or more factors.

```
mydata = read.csv('CompanyPayroll_9.25.csv')

A = mydata$Age
mydata$AgeGroup = ifelse(A < 23, 'Young',
                         ifelse(A > 30, 'Old', 'Middle-Aged'))

Y = mydata$Efficiency_Factor_Obs
xbarbar = mean(Y)
s2all = var(Y)

X1 = mydata$Job_Rank
X2 = mydata$AgeGroup

g1 = length(unique(X1))
g2 = length(unique(X2))
```

Null Hypotheses:

$H_0^{(1)}: \mu_1 = \mu_2 = \cdots = \mu_{g_1}$

$H_0^{(2)}: \mu_1 = \mu_2 = \cdots = \mu_{g_2}$

$H_0^{(1,2)}:$ There is no interaction effect between G_1 and G_2

Two-Factor ANOVA

Null Hypotheses:

$$H_0^{(1)}: \mu_1 = \mu_2 = \cdots = \mu_{g_1}$$
$$H_0^{(2)}: \mu_1 = \mu_2 = \cdots = \mu_{g_2}$$
$$H_0^{(1,2)}: \text{There is no interaction effect between } G_1 \text{ and } G_2$$

```
> SST
[1] 295.74
> SSM1
[1] 71.042
> SSM2
[1] 11.39
```

```
Associate.Middle-Aged
Associate.Old
Associate.Young
Ast. Manager.Middle-Aged
Ast. Manager.Old
Ast. Manager.Young
Entry.Middle-Aged
Entry.Old
Entry.Young
```

The metrics SST, SSM_1, and SSM_2 are defined exactly as they were in a (one-factor) ANOVA test.

Coding Recommendations: Usage of split(...) and sapply(...)

Since we are analyzing the variance with interaction, our SSE needs to consider the variance within intersectional groups.

```
Cells = split(Y, interaction(X1, X2, drop=T, lex.order=T))
```

$$SSE = \sum_{k=1}^{g_1 \cdot g_2} \left(\nu_k \cdot s_k^2\right)$$

```
varsp = function(S){
  if(length(S)>1){
    return(var(S))
  }else{
    return(0)
  }
}
```

Important: If $n_k \in \{0, 1\}$, take $s_k^2 := 0$.

```
> SSE
[1] 218.22
```

Two-Factor ANOVA

Null Hypotheses:

$H_0^{(1)}: \mu_1 = \mu_2 = \cdots = \mu_{g_1}$

$H_0^{(2)}: \mu_1 = \mu_2 = \cdots = \mu_{g_2}$

$H_0^{(1,2)}$: There is no interaction effect between G_1 and G_2

```
> SST
[1] 295.74
> SSM1
[1] 71.042
> SSM2
[1] 11.39
```

```
> SSE
[1] 218.22
```

The interaction-between-groups metric is given by:

$$SSM_{1,2} = \sum_{k=1}^{g_1 \cdot g_2} [n_k \cdot (\bar{x}_k - \bar{\bar{x}})^2]$$

```
> SSM12
[1] 77.523
```

Question: Does $SST = SSE + SSM_1 + SSM_2 + SSM_{1,2}$?

```
> SST
[1] 295.74
> SSE + SSM1 + SSM2 + SSM12
[1] 378.17
```

One “cosmetic” fix one could do is to renormalize for relative comparisons:

```
cc = SST/(SSE + SSM1 + SSM2 + SSM12)

SSM1r = SSM1*cc
SSM2r = SSM2*cc
SSM12r = SSM12*cc
SSEr = SSE*cc

SST
SSM1r + SSM2r + SSM12r + SSEr
```



```
> SST
[1] 295.74
> SSM1r + SSM2r + SSM12r + SSEr
[1] 295.74
```

Two-Factor ANOVA

Metric	Degrees of Freedom	Mean-Square Metrics
SST	$n-1$	$\frac{SST}{n-1}$
SSM_1^r	g_1-1	$\frac{\mathrm{SSM}_1^r}{g_1-1}$
SSM_2^r	g_2-1	$\frac{\mathrm{SSM}_2^r}{g_2-1}$
$\mathrm{SSM}_{1,2}^r$	$(g_1-1)\cdot(g_2-1)$	$\frac{\mathrm{SSM}_{1,2}^r}{(g_1-1)(g_2-1)}$
SSE^r	$n-(g_1\cdot g_2)$	$\frac{\mathrm{SSE}^r}{n-(g_1\cdot g_2)}$

```
cc = SST/(SSE + SSM1 + SSM2 + SSM12)

SSM1r = SSM1*cc
SSM2r = SSM2*cc
SSM12r = SSM12*cc
SSEr = SSE*cc

SST
SSM1r + SSM2r + SSM12r + SSEr
```

Our test statistics can then be defined as:

$$F_{stat}^{(1)} = \frac{MSM_1}{MSE} \quad \text{and} \quad F_{stat}^{(2)} = \frac{MSM_2}{MSE} \quad \text{and} \quad F_{stat}^{(1,2)} = \frac{MSM_{1,2}}{MSE}$$

each of which will follow their own F distribution as interpreted as usual.